Introduction to Calculus
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Differentiate the following expressions with respect to x.

a. $(3x^2 - 5x)^5$

b. $2x^2 + 5 - \frac{7}{x}$

c. $\frac{4x^3+5}{2x-9}$

(a) $\frac{d}{dx}(3x^2 - 5x)^5 = 5(3x^2 + 5x)^4(6x - 5)$ 1

(b) $\frac{d}{dx}(2x^2 + 5 - 7x^{-1}) = 4x + 7x^{-2} = 4x + \frac{7}{x^2}$ 1

(c) $\frac{d}{dx}\left(\frac{4x^3+5}{2x-9}\right) = \frac{(2x-9)(12x^2) - (4x^3+5) \times 2}{(2x-9)^2}$ 2

$$= \frac{24x^3 - 108x^2 - 8x^3 - 10}{(2x-9)^2}$$

$$= \frac{16x^3 - 108x^2 - 10}{(2x-9)^2}$$

2 Differentiate the following:

a. $2x^6 + 6x^4$ 1

b. $3\sqrt{x^3}$ 1

c. $-\frac{2}{x^3}$ 1

a) $\frac{d}{dx}(2x^6 + 6x^4) = 2 \times 6x^5 + 6 \times 4x^3 = 12x^5 + 24x^3$

b) $\frac{d}{dx}\left(3\sqrt{x^3}\right) = \frac{d}{dx}\left(3x^{\frac{3}{2}}\right) = 3 \times \frac{3}{2}x^{\frac{1}{2}}$

$= \frac{9}{2}x^{\frac{1}{2}} = \frac{9\sqrt{x}}{2}$

c) $\frac{d}{dx}\left(\frac{-2}{x^3}\right) = \frac{d}{dx}(-2x^{-3})$

$= -2 \times -3x^{-4} = 6x^{-4} = \frac{6}{x^4}$

3 The curve C has the equation $y = \frac{1}{3}x^3 - 4x^2 + 8x + 3$. The point P has coordinates $(3, 0)$.

a. Show that P lies on C. 1

b. Find the equation of the tangent to C at P. 2

c. Another point Q also lies on C. The tangent to C at Q is parallel to the tangent to C at P. What are the coordinates of Q? 2

(a) Substitute P into the equation of C

$y = \frac{1}{3} \times 3^3 - 4 \times 3^2 + 8 \times 3 + 3 = 0$

$\therefore$ P lies on C

(b) $\frac{dy}{dx} = x^2 - 8x + 8$

When $x = 3$ $\quad m = 3^2 - 8 \times 3 + 8 = -7$

Equation of the tangent

$y - y_1 = m(x - x_1)$

$y - 0 = -7(x - 3)$

$y = -7x + 21$

(c) Tangent at Q is parallel and has the same gradient ($m = -7$)

$\frac{dy}{dx} = x^2 - 8x + 8 = -7$

$x^2 - 8x + 15 = 0$

$(x - 3)(x - 5) = 0$

$x = 3$ or $x = 5$

The x-coordinate of Q is 5

$y = \frac{1}{3} \times 5^3 - 4 \times 5^2 + 8 \times 5 + 3 = -15\frac{1}{3}$

$\therefore$ Point Q is $\left(5, -15\frac{1}{3}\right)$

4 A particle moves along the x-axis such that its displacement from the origin (in cm) at a time t (seconds) is given by the equation: $x = 2t^3 + \sqrt{t}$. Find the velocity and acceleration of the particle after 9 seconds. 2

$x = 2t^3 + t^{\frac{1}{2}}$

$\dot{x} = 6t^2 + \frac{1}{2}t^{-\frac{1}{2}}$

$\ddot{x} = 12t - \frac{1}{4}t^{-\frac{3}{2}}$

When $t = 9$

$\dot{x} = 6(9)^2 + \frac{1}{2}(9)^{-\frac{1}{2}}$

$\dot{x} = 6(9)^2 + \frac{1}{2} \times \frac{1}{\sqrt{9}}$

Velocity $= 486\frac{1}{6}$

When $t = 9$

$\ddot{x} = 12(9) - \frac{1}{4}(9)^{-\frac{3}{2}}$

$= 108 - \frac{1}{4} \times \frac{1}{\sqrt{9^3}}$

$= 108 - \frac{1}{108} = 107\frac{107}{108}$

5 A water tank is being filled at a variable rate. The height of the water, h cm, at any time, t minutes can be described by $h(x) = 2t^2 - 4t + 50$. Find:

a. the initial height of water in the tank. 1

b. the instantaneous rate at which the height is changing at 4 minutes. 1

c. the average rate at which the height has changed over the first 4 minutes. 2

(a)	Initially $t = 0$ $h(0) = 2 \times 0^2 - 4 \times 0 + 50 = 50$ cm $\therefore$ Initial height is 50 cm
(b)	Use the derivatives to find the instantaneous rate of change $h'(t) = 4t - 4$ $h'(4) = 4 \times 4 - 4 = 12$ $\therefore$ Instantaneous rate is 12 cm/min
(c)	To find the height when $t = 4$ $h(4) = 2 \times 4^2 - 4 \times 4 + 50 = 66$ cm $\text{Average rate} = \frac{\text{Total change in depth}}{\text{Total time taken}} = \frac{66 - 50}{4} = 4 \text{ cm/min}$ $\therefore$ Average rate is 4 cm/min

6 Use the definition: $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, to show that the derivative of $f(x) = 3x^2 + x$ is $6x + 1$. 3

$$f(x) = 3x^2 + x \qquad f(x+h) = 3(x+h)^2 + (x+h)$$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{3(x+h)^2 + (x+h) - 3x^2 - x}{h} \\ &= \lim_{h \to 0} \frac{3(x^2 + 2xh + h^2) + (x+h) - 3x^2 - x}{h} \\ &= \lim_{h \to 0} \frac{3x^2 + 6xh + 3h^2 + x + h - 3x^2 - x}{h} \\ &= \lim_{h \to 0} \frac{6xh + 3h^2 + h}{h} = \lim_{h \to 0} 6x + 3h + 1 = 6x + 1 \end{aligned}$$

7 Find the derivative of:

a. $\frac{x^2+1}{x+1}$ 1

b. $(x^3 + 2)^5$ 1

c. $x^7(x^4 + 5x)^6$ 2

a)	$\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \frac{(x+1)(2x) - (x^2+1)(1)}{(x+1)^2} = \frac{2x^2 + 2x - x^2 - 1}{(x+1)^2} = \frac{x^2 + 2x - 1}{(x+1)^2}$
b)	$\frac{d}{dx}((x^3+2)^5) = 5(x^3+2)^4(3x^2) = 15x^2(x^3+2)^4$
c)	$\frac{d}{dx}(x^7(x^4+5x)^6) = x^7(6(x^4+5x)^5(4x^3+5)) + 7x^6(x^4+5x)^6$ $= (24x^{10} + 30x^7)(x^4+5x)^5 + 7x^6(x^4+5x)^6$ Can be further simplified but acquires no extra marks. $= (x^4+5x)^5(24x^{10} + 30x^7 + 7x^{10} + 35x^7)$ $= (x^4+5x)^5(31x^{10} + 65x^7)$

8 A point $P(-2,-8)$ lies on the curve $y = 3x^3 + 4x^2$.

a. Find the equation of the tangent to the curve at P. 2

$$\begin{aligned} y &= 3x^3 + 4x^2 \\ y' &= 9x^2 + 8x \\ \text{at } x &= -2 \\ y' &= 36 - 16 = 20 \\ \therefore m_1 &= 20 \end{aligned}$$

$$\begin{aligned} y - -8 &= 20(x - -2) \\ y + 8 &= 20x + 40 \\ y &= 20x + 32 \\ 20x - y + 32 &= 0 \end{aligned}$$

b. Find the equation of the normal to the curve at P. 2

$$\begin{aligned} m_2 &= -\frac{1}{20} \\ y + 8 &= -\frac{1}{20}(x + 2) \\ -20y - 160 &= x + 2 \\ x + 20y + 162 &= 0 \\ y &= -\frac{x}{20} - \frac{81}{10} \end{aligned}$$

9 If $y = f\left(\frac{1}{x}\right)$ which of the following is $\frac{dy}{dx}$? 1

A. $f'\left(\frac{1}{x}\right)$

B. $-\frac{1}{f(x)}$

C. $-\frac{f'\left(\frac{1}{x}\right)}{x^2}$

D. $f'\left(-\frac{1}{x^2}\right)$

$$\frac{dy}{dx} = f'\left(\frac{1}{x}\right) \times \left(\frac{1}{x}\right)' = -\frac{1}{x^2} f'\left(\frac{1}{x}\right)$$

Introduction to Calculus
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 A tangent is drawn to the curve $y = 2x^3 + 3x^2$ at $x = 1$. What is the gradient of the tangent? 1

$$y'(1) = 12$$

2 Differentiate, with respect to x:

a. $\frac{7x^3-4}{x+1}$ 2

b. $\frac{5}{\sqrt{2x-1}}$ 1

a) $y = \frac{7x^3-4}{x+1}$ $u = 7x^3 - 4$, $u' = 21x^2$, $v = x+1$, $v' = 1$

$$y' = \frac{21x^2(x+1) - (7x^3-4)}{(x+1)^2}$$

$$= \frac{21x^3 + 21x^2 - 7x^3 + 4}{(x+1)^2}$$

$$= \boxed{\frac{14x^3 + 21x^2 + 4}{(x+1)^2}}$$

b). $y = \frac{5}{\sqrt{2x-1}} = 5(2x-1)^{-\frac{1}{2}}$

$$y' = 5 \times -\frac{1}{2} \times 2(2x-1)^{-\frac{3}{2}} = \boxed{-\frac{5}{\sqrt{(2x-1)^3}}}$$

3 A particle moves so that its displacement from the origin is $x = t^3 - 9t$ where x is displacement in metres and t is time in seconds. Find:

a. When and where the particle is at rest. 2

b. The acceleration at the time determined in (i). 1

(i) $x = t^3 - 9t$ $\quad v = \frac{dx}{dt} = \dot{x} = 3t^2 - 9$

Particle at rest when $v = 0$

$\therefore 3t^2 - 9 = 0 \quad \therefore t^2 = 3$

$\therefore t = \sqrt{3}$ seconds $(t > 0)$

@ $t = \sqrt{3}$, $x = (\sqrt{3})^3 - 9.\sqrt{3} = 3\sqrt{3} - 9\sqrt{3} = -6\sqrt{3}$ metres

(ii) $\ddot{x} = \frac{d^2x}{dt^2} = \frac{dv}{dt} = 6t$

@ $t = \sqrt{3}$, $\ddot{x} = \frac{d^2x}{dt^2} = \frac{dv}{dt} = 6\sqrt{3}\ ms^{-2}$

4 a. Find the equation of the tangent to the curve $y = (2x-3)^3$ at the point where $x = 1$. 3

b. Find the angle of inclination of this tangent line correct to the nearest degree. 1

a. $y = (2x-3)^3$

$\frac{dy}{dx} = 6(2x-3)^2$

$x = 1 \Rightarrow \begin{cases} y = -1 \\ \frac{dy}{dx} = 6 \end{cases}$

Tangent at $(1, -1)$ has gradient 6

and equation $y + 1 = 6(x-1)$

$y = 6x - 7$

b. $\tan\theta = 6 \qquad \therefore \theta \approx 81°$

5 Use the formula $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 + 3$. 2

$f(x+h) = (x+h)^2 + 3$

$f(x) = x^2 + 3$

$f(x+h) - f(x) = h(2x+h)$

$$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{h(2x+h)}{h}$$

$$= \lim_{h\to 0}(2x+h)$$

$$= 2x$$

$\therefore f'(x) = 2x$

6 What is the gradient of the normal at $x = -1$ when $y = 3e^x - ex^3$? 3

$y' = 3e^x - 3ex^2$

when $x = -1$, $y' = 3e^{-1} - 3e(-1)^2$

$m_T = \frac{3}{e} - 3e = \frac{3 - 3e^2}{e}$

$\therefore \boxed{m_N = -\frac{e}{3(1-e^2)}}$

7 What is $\frac{d}{dx}\left(\sqrt{1+x^2}\right)$. $\frac{d}{dx}\left[(1+x^2)^{\frac{1}{2}}\right] = \frac{1}{2} \times 2x\,(1+x^2)^{-\frac{1}{2}} = \frac{x}{\sqrt{1+x^2}}$ 1

8 Consider the product function $v(x) = f\big(g(x)\big)h(x)$. What is $\frac{dv}{dx}$? 1

A. $g'(x)h(x) + f\big(g(x)\big)h'(x)$

B. $f'\big(g(x)\big)h(x) + f\big(g(x)\big)h'(x)$

C. $f'\big(g(x)\big)g'(x)h(x) + f\big(g(x)\big)h'(x)$

D. $f'\big(g(x)\big)h'(x) + f\big(g(x)\big)h(x)$

$\frac{dv}{dx} = f[g(x)]\,h'(x) + g'(x)\,f'[g(x)]\,h(x)$

C

9 A water tank is being filled at a variable rate. The height of the water, h cm, at any time, t minutes can be described by $h = t^3 + 2t^2 - 4t + 50$. Find the instantaneous rate at which the height is changing at 4 minutes. 2

$h = t^3 + 2t^2 - 4t + 50$

$\frac{dh}{dt} = 3t^2 + 4t - 4$

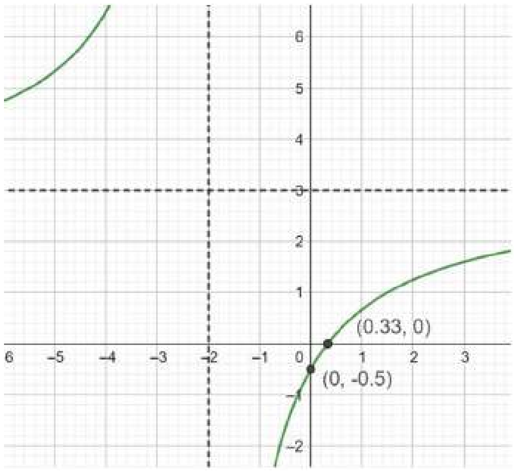
When $t = 4, \frac{dh}{dt} = 3.4^2 + 4.4 - 4 = 60$ cm / min

10 Consider the function $f(x) = \frac{3x-1}{x+2}$.

$$\frac{3x-1}{x+2} = \frac{3(x+2)-7}{x+2} = 3 - \frac{7}{x+2}$$

a. Show that $f(x) = 3 - \frac{7}{x+2}$. 1

b. Sketch the graph of $y = f(x)$, indicating asymptotes and intercepts in the space below. 3



c. Find $f'(x)$. 2

$$\left(3 - 7(x+2)^{-1}\right)' = 7(x+2)^{-2} = \frac{7}{(x+2)^2}$$

d. Let $g(x) = x - 3$. Find all values of x such that the tangent line to $f(x)$ is parallel to $g(x)$. 2

$$f'(x) = 1 \Rightarrow \frac{7}{(x+2)^2} = 1 \Rightarrow x = -2 \mp \sqrt{7}$$

e. Find the equation of the line that is perpendicular to the line $y = g(x)$ and passes through the point $(2, -5)$. 2

$$m = -1 \Rightarrow y + 5 = -1(x-2) \Rightarrow y = -x - 3$$